04bis. Task analysis

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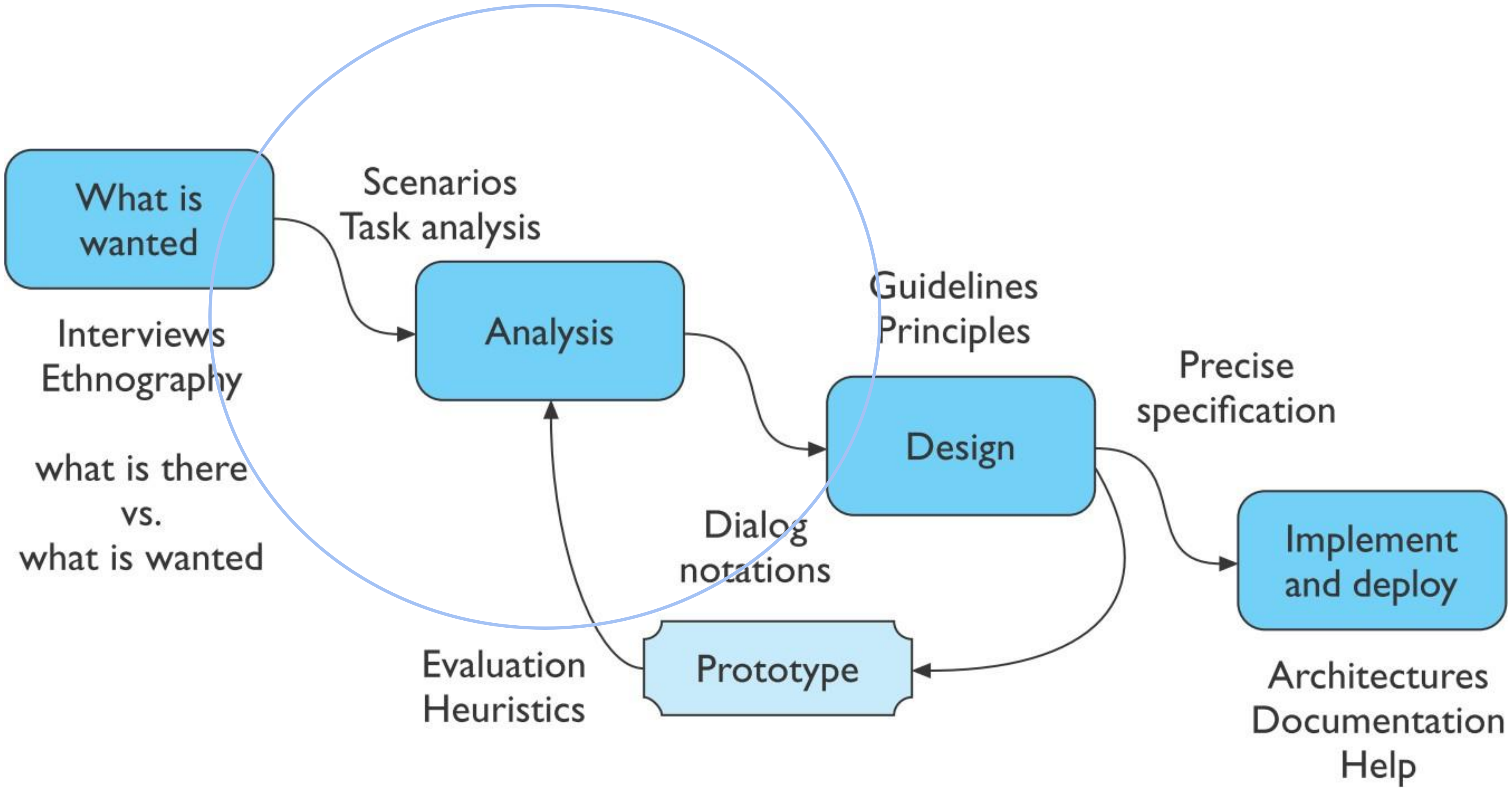
Human-Centered Design Process



Summary

- Create **design needs/goals**
  - As an intermediate representation before the user interface design
- Make the user needs' analysis **explicit**
- Think about the *interplay* between the activity that someone has and the interface we offer
- **Represent** and synthetize the results of the analysis and the design goals
- They are different from user needs…
- …we make a simplification and we consider that needs=goals

Hall of Fame or Shame?

User Needs Edition

Beware: we are missing the general context here!

- Users need a faster horse
- Users need to have financial help
- Users needs a way to move faster from one place to another
- Users need to have more tools
- Users need to practice more with the appropriate tools
- Users need to be able to run faster

"Needs are human emotional or physical necessities. [...]
Needs are verbs (activities and desires with which your user could use help), not nouns (solutions). [...]
It can be helpful to use the phrases 'needs a way to' or 'needs to be able to' in your list of user needs."

Needs/Goals

- Needs drive Goals: needs are the driving force behind goals
  - e.g.: to improve well-being (need), set the goal of exercising regularly, monitoring sleep, or eating healthier
- Multiple Goals can derive from a single Need
- Goals translate Needs into actions: goals are a concrete form of needs
- Context influences Goals, but not Needs: goals vary depending on the context, available resources, but the underlying needs remain the same
  - e.g.: need is "feeling secure"; goal can be "getting a promotion at work" or "installing a security system at home"
- **but we make a simplification: Needs = Goals**

Task Analysis

- (Need/Goal-Centered) Task Analysis is the study of the way people perform their activities
- Aim is to determine:
  - what they **do** (steps)
  - what things they **use** (artifacts)
  - how well they **succeed** (goals)

Sample Task: To Clean The House (I)

- **Steps**:
  - get the vacuum cleaner out
  - fix the appropriate attachments
  - clean the rooms
  - when the dust bag gets full, empty it
  - put the vacuum cleaner and tools away
- Must know and use different **artifacts**:
  - vacuum cleaners, their attachments, dust bags
  - cupboards, rooms
  - …

Sample Task: To Clean The House (II)

- **Goals:**
  - Here your *point of view* comes in
  - Removing dust? -> narrow goal
  - Tidying up the house after a party?
  - Hosting people for the dinner?
  - Having a satisfying evening? -> wide goal

Another Example of Task (with Steps)

- A person preparing an overhead projector for use would be seen to carry out the following steps:

1. Plug in to main and switch on supply.
2. Locate on/off switch on projector.
3. Discover which way to press the switch.
4. Press the switch for power.
5. Put on the slide and orientate correctly.
6. Align the projector on the screen.
7. Focus the slide.

What is a Tasks?

- «A **task** is a **goal** together with some ordered set of **actions**.» (Benyon)

Goal

- A state of the application domain that a work system (user+technology) wishes to achieve.
- Specified at particular levels of abstraction.

Task

- A structured set of activities required, used, or believed to be necessary by an agent (human, machine) to achieve a goal using a particular technology.
- The task is broken down into more and more detailed levels of description until it is defined in terms of actions.

Action

- An action is a task that has no problem solving associated with it and which does not include any control structure.
- Actions are ‘simple tasks’.

What You Learn with Task Analysis

- What your users’ goals can be; what they are trying to achieve
- What users actually do to achieve those goals
- What experiences (personal, social, and cultural) users bring to the tasks
- How users are influenced by their physical environment
- How users’ previous knowledge and experience influence:
  - How they think about their work
  - The workflow they follow to perform their tasks
  - The pain points they experience to perform the tasks

Why Is It Useful?

- Task analysis is the process of learning about ordinary users by observing them in action to understand in detail **how** they perform their **tasks** and achieve their intended **goals**

  seldom expressed by users

- Tasks analysis helps in:
  - Identifying the tasks that your application **must support**
  - Refining or re-defining your app's navigation or search
  - Application requirements gathering
  - Developing your content strategy and app structure
  - The initial stages of Prototyping
  - Performing usability testing

Examples

- example needs/goals:
  - monitor my health
  - better organize my time
- derived tasks:
  - entering health data into a digital tracking system
  - setting reminders for daily exercise
- poorly defined tasks:
  - improve diet (not operational, too vague)
  - use the medical app (too generic and tied to a specific interface)
- derived actions:
  - measure your health parameters, such as heart rate or blood pressure, using a tool or device that is convenient for you
  - record the data in a way that allows for easy review later (e.g., writing it in a log, using a physical or digital tracking system)

Characteristics of Task Analysis

- Task analysis is easier when you have well-defined **workflows** (e.g., planning a trip somewhere)
  - or **repeated activities**, such as scheduling
- Challenge:
  - We **do not** design tasks, but interfaces
  - Tasks and objects do not map 1:1
    - e.g., a web app has multiple tasks
  - People use the same interface and application to achieve slightly different results or do things differently one another

Example of needs

- Service: Uber
- Need for **convenient transportation**
  - The user wants a quick and easy way to get from point A to point B without the hassle of finding public transportation or driving themselves
- Need for **safety during travel**
  - The user wants to feel safe while traveling, whether through vetted drivers, real-time tracking, or ensuring someone knows their location
- Need for **cost-effective transportation**
  - The user wants to balance convenience with affordability and ensure they get a reasonably priced ride

Example of tasks (Need-Centered task Analysis)

- Need: **convenient transportation**
  - Task: arrange transportation from one location to another; the app allows the user to request a ride by defining their starting point and destination, and it matches them with a nearby driver
- Need: **safety during travel**
  - Task: verify driver details and monitor the ride's progress; the app provides information about the driver, such as their identity and vehicle details, and offers the ability to track the vehicle's real-time location during the trip
- Need: **cost-effective transportation**
  - Task: evaluate transportation options based on cost and availability; the app allows the user to compare different transportation options (e.g., shared rides, private rides) and provides fare estimates, helping the user make a decision based on their budget

Exercise

- Service: Glovo/JustEat
- Needs: ?
- Tasks: ?

References

- Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale: Human Computer Interaction, 3rd Edition, Chapter 15 "Task Analysis"
- David Benyon: Designing Interactive Systems, Chapter 11 "Task Analysis"
- http://www.usabilitybok.org/task-analysis
- https://www.usability.gov/how-to-and-tools/methods/task-analysis.html

Attribution

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  - http://www.inf.ed.ac.uk/teaching/courses/hci/0708/lecs/tasks.pdf
  - https://www.tutorialspoint.com/human_computer_interface/design_process_and_task_analysis.htm
  - https://www.slideshare.net/alanjohndix/hci-3e-ch-15-task-analysis

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